Prof. Dhanapalan College
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Project Document

EduGenei

AI Study Plan Generator using Gemini Models

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Submitted by

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Abstract

EduGenie is an AI-powered educational web application designed to create personalized 7-day study plans for users with the assistance of Google Gemini AI models. The user creates an account and specifies key parameters such as target subject, academic goals, learning pace, and available study hours. The server processes these details using Gemini AI models to generate customized, dynamic study blueprints, interactive summaries, and practice recommendations to enhance academic performance.

1. Objectives

- **Personalized Learning Architecture:** To develop an intelligent web platform that provides tailored learning resources and structured schedules using Google Gemini AI models.
- **Goal & Progress Tracking:** To enable users to set custom academic targets, generate adaptive 7-day study blueprints, and systematically track topic completion.
- **Interactive Knowledge Synthesis:** To simplify complex educational subjects through dynamic AI-generated explanations, topic breakdowns, and targeted practice content.
- **Seamless User Experience:** To provide an intuitive, responsive interface with robust account management to streamline daily study workflows.

2. Scope of the Project

The scope of **EduGenie** encompasses the design, development, and deployment of a personalized AI study companion tailored for modern learners:

- **Custom Study Blueprint Generation:** Automatically formulates tailored 7-day study timelines based on subject complexity, user availability, and individual learning velocity.
- **Generative AI Integration:** Utilizes Google Gemini API to dynamically process user queries, output organized summary guides, and break down complex concepts.
- **Profile & Preference Management:** Securely maintains user credentials, saved subjects, target milestones, and historical learning performance.
- **Responsive & Cross-Platform UI:** Ensures seamless usability across desktop computers, tablets, and mobile browser environments.

3. System Study

3.1 Existing System

In traditional academic setups, students rely on static physical timetables, generic study guides, or manual schedule planning. Key drawbacks include:

- **High Manual Effort:** Creating organized study plans manually consumes significant time and effort.
- **Lack of Personalization:** Pre-defined schedules fail to adapt to individual strengths, weaknesses, or unexpected delays.
- **Static Content:** Learning material cannot dynamically adjust when a student requires deeper explanation on specific topics.

3.2 Proposed System

The proposed system, **EduGenie**, overcomes these limitations by integrating intelligent AI capabilities:

- **Automated Customization:** Employs Google Gemini AI models to build personalized schedules in seconds.
- **Dynamic Adaptation:** Allows students to generate instant concept explanations, custom quizzes, and tailored revision schedules.
- **Centralized Hub:** Merges authentication, progress monitoring, and study planning into a unified web application.

4. System Requirements

4.1 Software Requirements

Item	Details
Language	Python 3.11 or newer
Framework	Flask / Django
AI API	Google Gemini API
Frontend	HTML5, CSS3, JavaScript
Database	SQLite / PostgreSQL
Operating System	Windows 10/11, macOS, or Linux

4.2 Hardware Requirements

Item	Details
Processor	Intel Core i3 / AMD Ryzen 3 or higher
RAM	4 GB minimum (8 GB recommended)
Storage	256 GB SSD / HDD
Network	Active Internet Connection (for Gemini API calls)

5. System Design & Database Schema

5.1 Database Design

User Table

Field	Data Type	Constraint	Description
`user_id`	Integer	Primary Key, Auto Inc.	Unique identifier for user
`username`	Varchar(50)	Unique, Not Null	Account username
`email`	Varchar(100)	Unique, Not Null	Registered user email
`password_hash`	Varchar(255)	Not Null	Encrypted password string

Study Plan Table

Field	Data Type	Constraint	Description
`plan_id`	Integer	Primary Key, Auto Inc.	Unique plan identifier
`user_id`	Integer	Foreign Key	References `User.user_id`
`subject`	Varchar(100)	Not Null	Target subject or course
`plan_content`	Text	Not Null	Generated study schedule JSON

6. Use of Gemini Models

Google Gemini AI serves as the core reasoning engine for **EduGenie**. The model handles complex textual synthesis and structures recommendations based on targeted prompt engineering:

- **Structured Plan Generation:** Converts user parameters (subject, target hours, goal) into clean, JSON-formatted 7-day study routines.
- **Adaptive Explanations:** Dynamically elaborates on specific sub-topics whenever a student requests clarification.
- **Practice Content Creation:** Generates topic-specific quizzes and review flashcards to reinforce retention.

7. Security & Deployment

- **API Key Security:** Google Gemini API keys are isolated using server environment variables (`.env`) to prevent source exposure.
- **Authentication & Hashing:** Passwords are encrypted using industry-standard `bcrypt` hashing algorithms prior to storage.
- **Sanitization & Validation:** All frontend user inputs are validated to protect against SQL Injection and Cross-Site Scripting (XSS).

8. System Testing

- **Unit Testing:** Individual route handlers, password encryption methods, and API request formatters were validated independently.
- **Integration Testing:** Verified seamless communication between the frontend interface, Python backend, database, and Gemini API endpoints.
- **Responsive UI Testing:** Confirmed consistent presentation across desktop and mobile screens.

9. Limitations & Future Enhancements

> **Current Limitations:** Requires active internet connectivity for Gemini API processing; currently focuses primarily on structured text and Markdown outputs without direct video or real-time audio generation.

- **Multimodal Support:** Integrate image recognition to analyze uploaded textbook pages and handwritten notes.
- **Gamification & Badges:** Add reward badges, progress streaks, and achievement levels to motivate daily study habits.
- **Institutional LMS Integration:** Enable seamless schedule syncing with platforms like Google Classroom, Canvas, and Moodle.

10. Conclusion

EduGenie successfully demonstrates the integration of modern Artificial Intelligence in personal education and task management. By leveraging Google Gemini AI models within a lightweight web framework, the system provides automated, highly personalized 7-day study schedules tailored to individual learning paces and goals.

The application resolves the static, rigid limitations of traditional study planning by offering dynamic, structured, and easily manageable study blueprints. While current scope limitations exist around external LMS sync and live video tutoring, **EduGenie** establishes a robust framework for next-generation, AI-assisted learning tools.

11. References

1. **Google AI for Developers.** *Gemini API Documentation and Python SDK Reference.* Google, https://ai.google.dev/docs
2. **Flask Documentation.** *Web Development in Python.* Pallets Projects, https://flask.palletsprojects.com/
3. **Python Software Foundation.** *Python 3.11 Standard Library and Language Reference.* https://docs.python.org/3/
4. **MDN Web Docs.** *HTML5, CSS3, and JavaScript Guides for Responsive Web Design.* Mozilla Developer Network, https://developer.mozilla.org/
5. **PostgreSQL / SQLite Global Development Group.** *Relational Database Systems Documentation.*